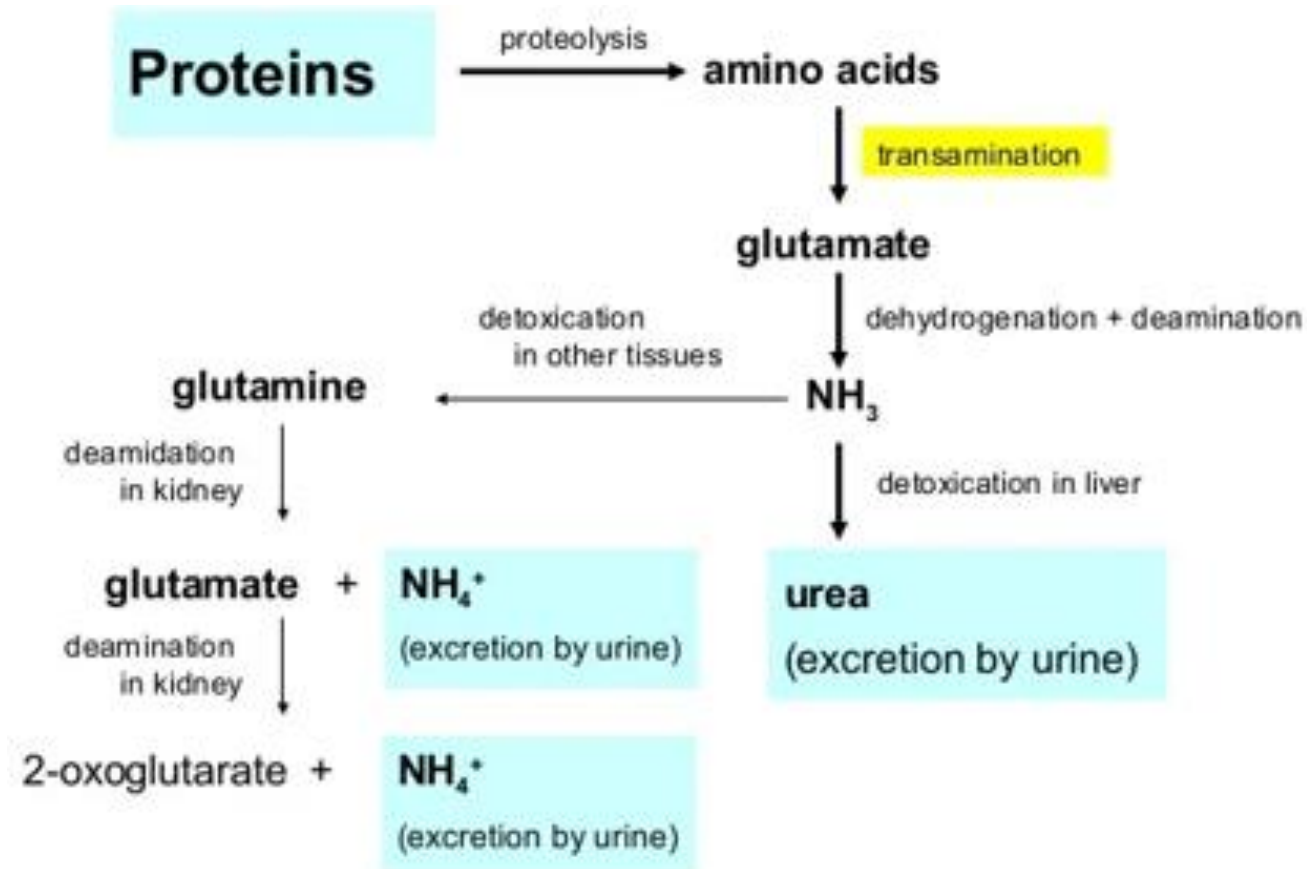


Utilisation of Proteins

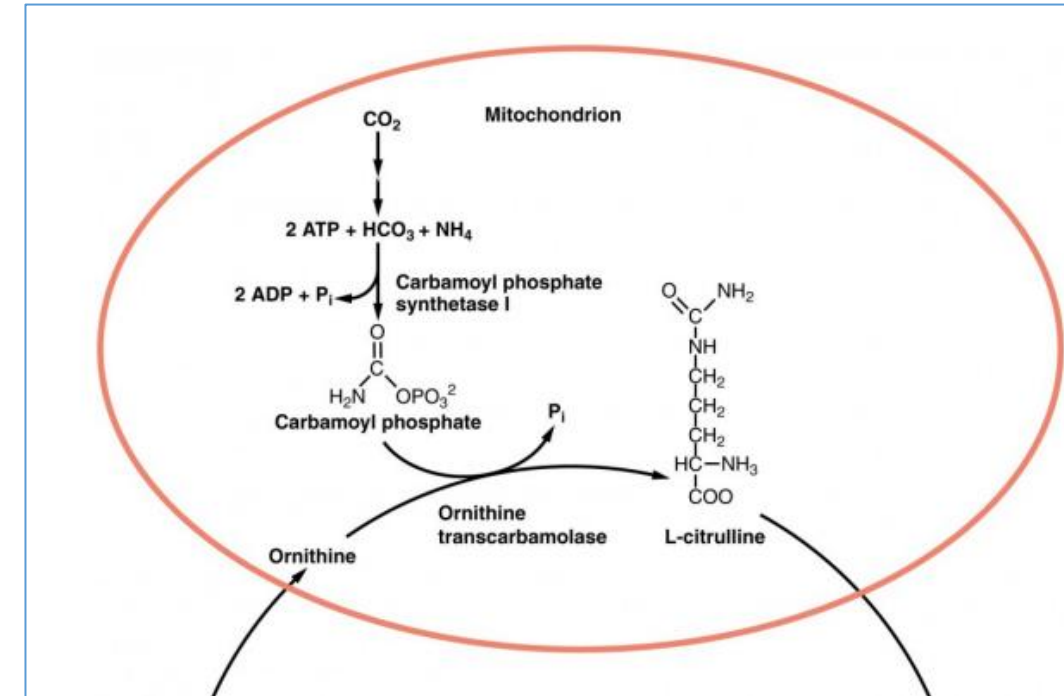
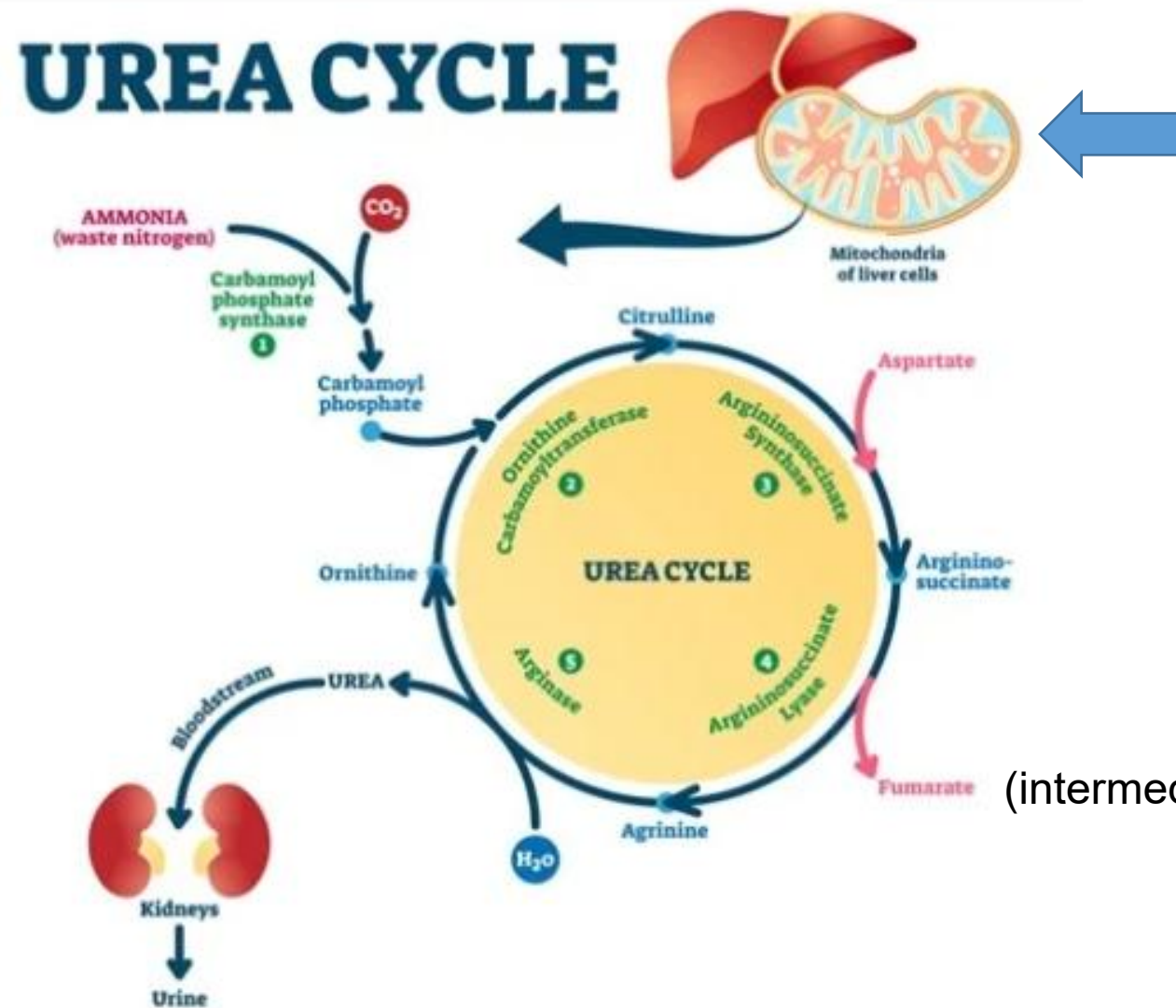
- Proteins are everywhere in the living system and are in many forms
- As enzymes, signalling molecules, transporters, in structures of bones and muscles
- Proteins are not stored for later use and excess protein is always **converted into glucose or triglycerides or ketones or decomposed**
- Amino acids are used to make new proteins or converted into fats or acetyl-CoA and enter into the Krebs cycle
- Amino acid decomposition results in hydrocarbons and nitrogenous wastes (Urea Cycle)
- **Urea cycle** occurs in the **liver** and to some extent in the **kidney**
- Products of the urea cycle is also linked to the Krebs cycle
- **Deamination** also occurs when excess of amino acids are present; this is converted into ammonia
- Ammonia is produced by the degradation of amino acids or gut bacteria
- Excess of ammonia in blood is toxic
- Urea cycle removes ammonia from blood and is converted to urea, and eliminated ultimately

Utilisation of Proteins – Deamination and Transamination

Deamination and transamination reactions



Utilisation of Proteins – Urea Cycle



(intermediate product in the Krebs cycle)

Utilisation of Proteins – Ketogenic and Glucogenic Amino Acids

Glucogenic and ketogenic amino acids

Glucogenic amino acids:

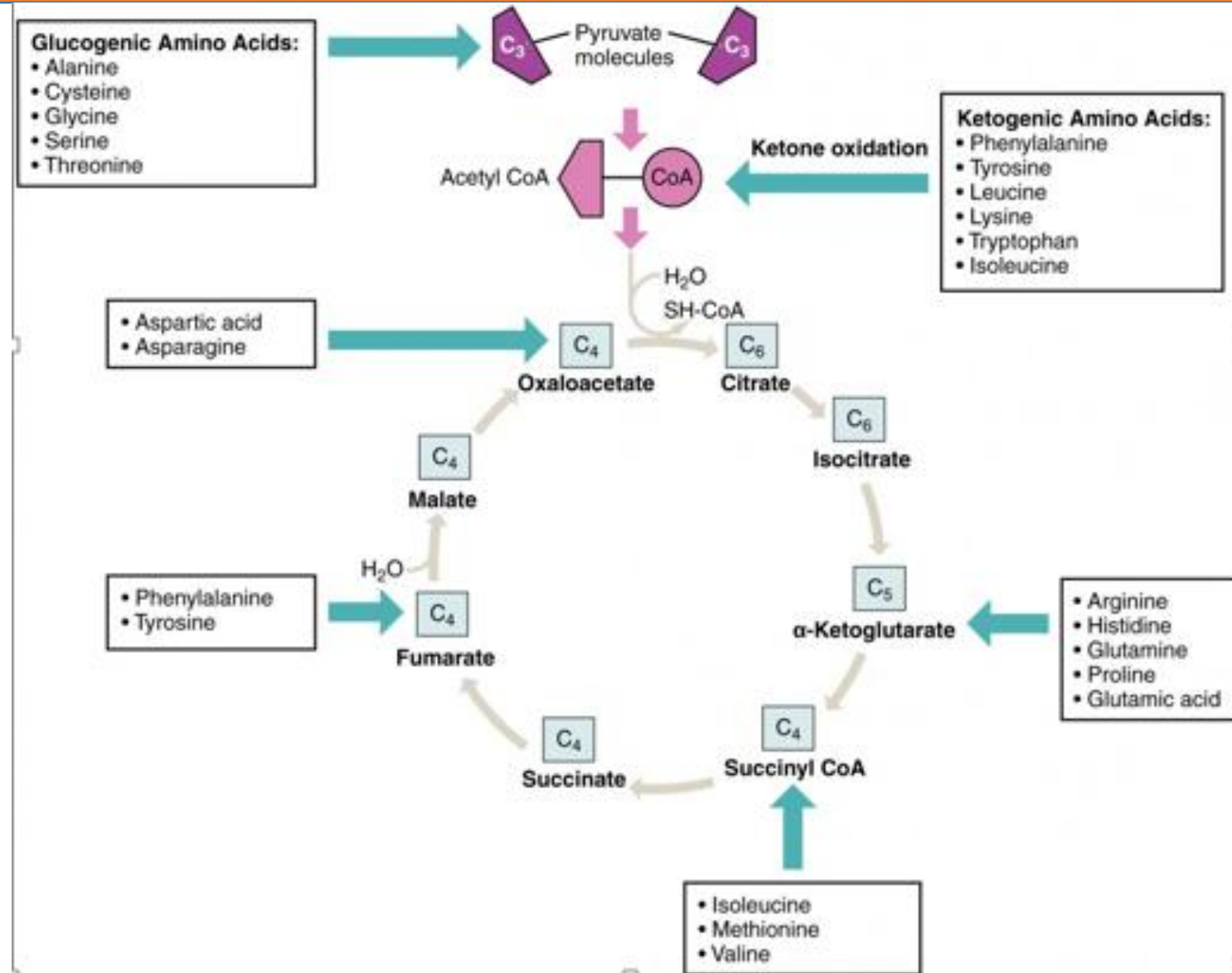
Amino acids which produce pyruvate or any other glucose precursors

Ketogenic amino acids:

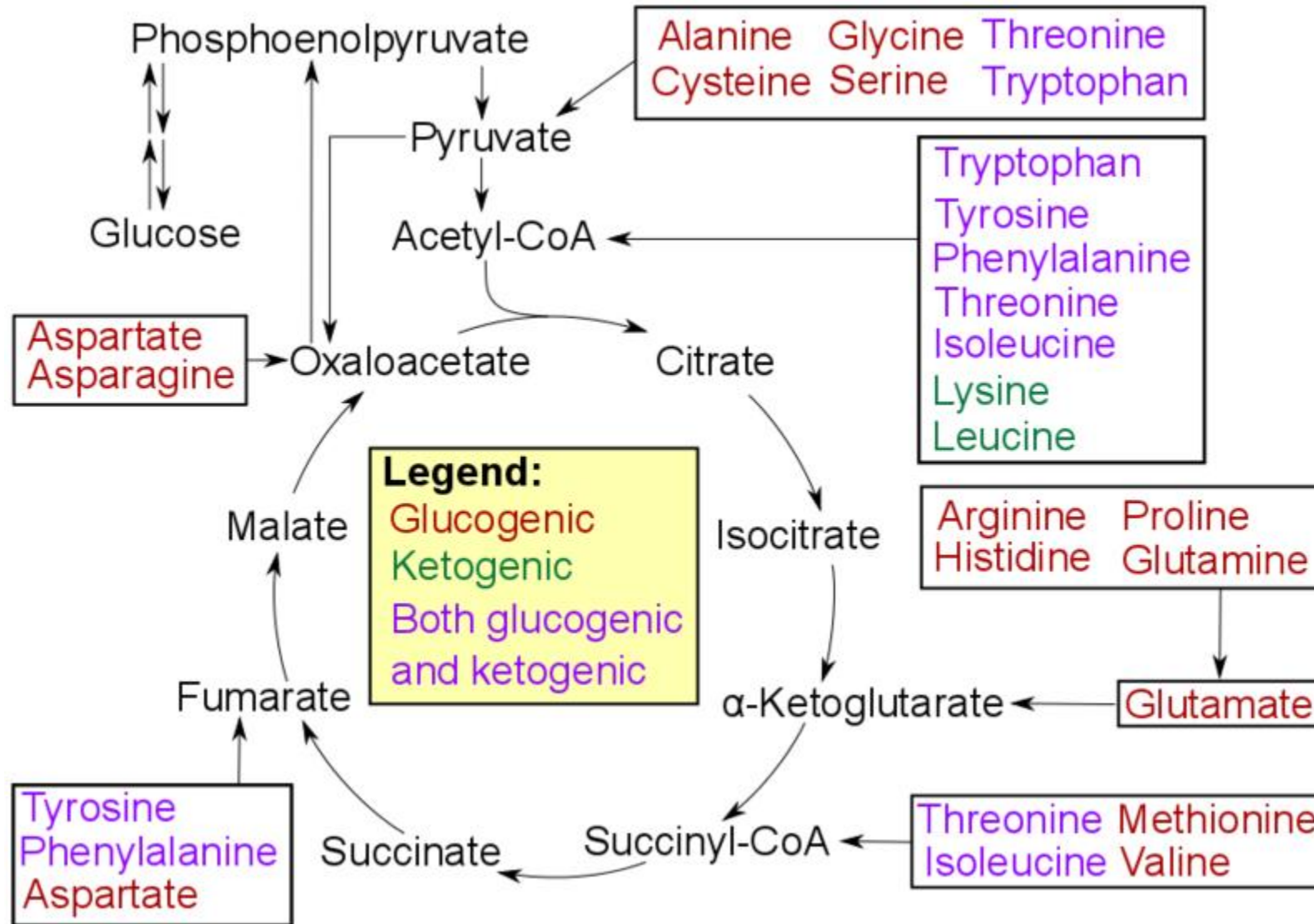
Amino acids which produce acetyl CoA or acetoacetyl CoA

Some amino acids are just ketogenic while some are glucogenic; there are amino acids which are both ketogenic and glucogenic

Amino acids can also enter the Krebs cycle and contribute to energy production



Glucogenic and Ketogenic Amino Acids

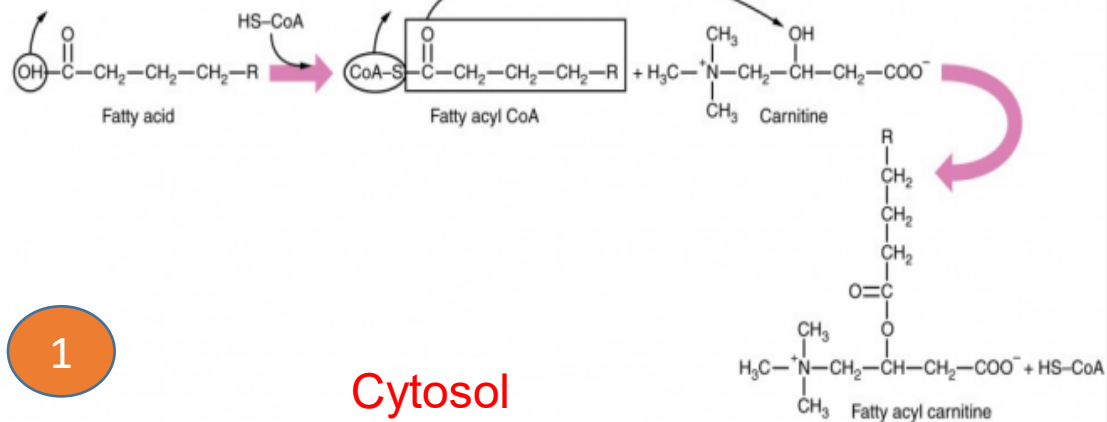


Utilisation of Lipids

- Fats are ingested or synthesised by the body by **adipocytes or hepatocytes** from carbohydrate precursors
- Lipids in circulation, as chylomicrons, either go to the liver or stored in fat cells (adipocytes)
- Lipid metabolism involved oxidation of fatty acids either to generate energy or to synthesize new lipid molecules
- Lipid metabolism is associated with carbohydrate metabolism
- When glucose level is low, triglycerides are converted to acetyl CoA and used to generate ATP through aerobic respiration
- **Fatty acid oxidation (or beta oxidation):** Breakdown of fatty acids to produce fatty acyl CoA (in cytoplasm)
- Transported to mitochondria where it forms acetyl CoA and then enters the **Krebs cycle**

Utilisation of Lipids – Fatty acid oxidation

1) Converting a fatty acid to fatty acyl carnitine allows transport through the mitochondrial membranes.

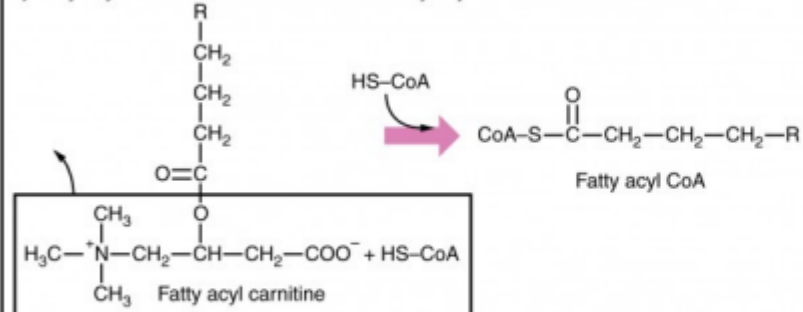


Cytosol

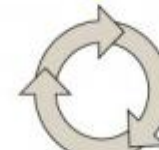
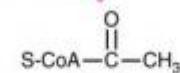
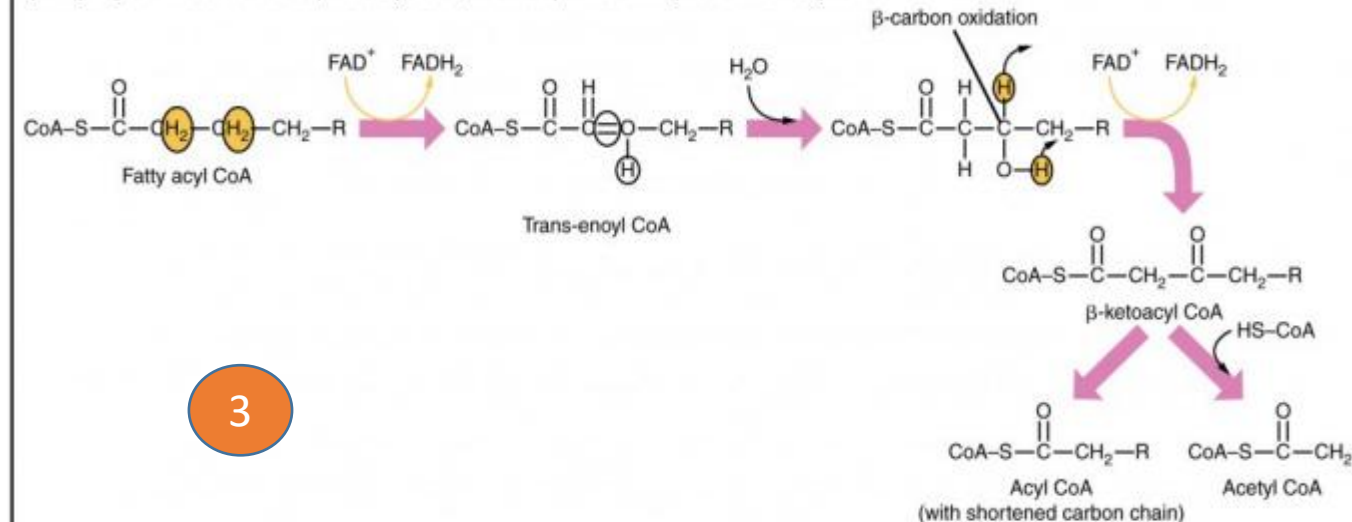
Fatty acyl carnitine enters mitochondrial matrix



2) Fatty acyl carnitine is converted back to fatty acyl CoA within a mitochondrion.

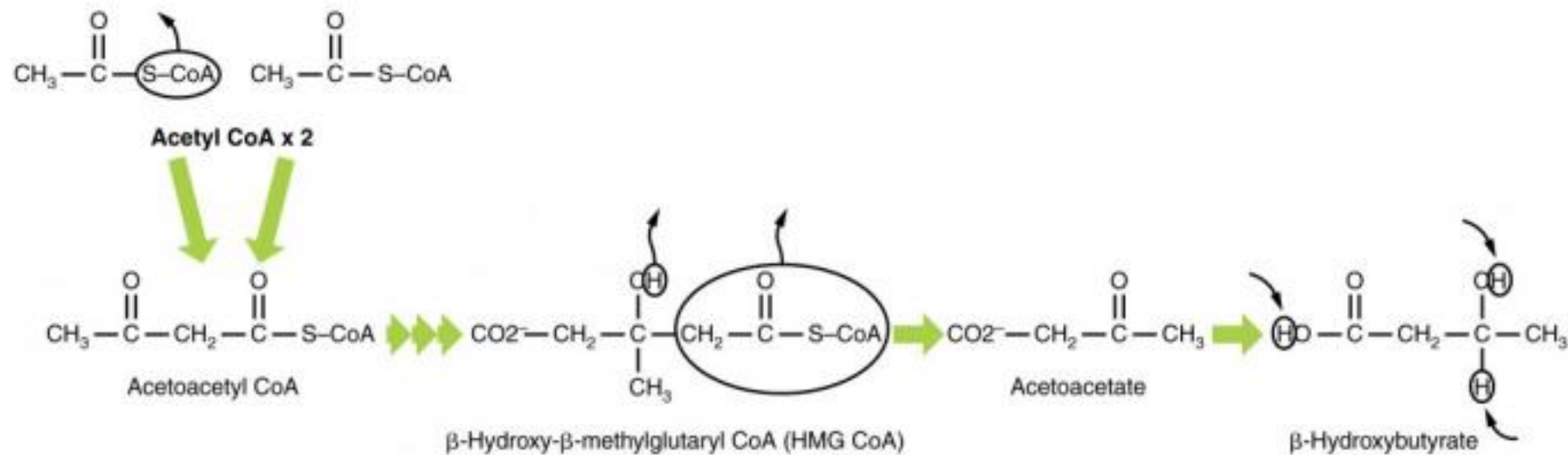


3) Fatty acyl CoA is converted to β -ketoacyl CoA, which is split into an Acyl CoA and Acetyl CoA



Utilisation of Lipids – Ketogenesis

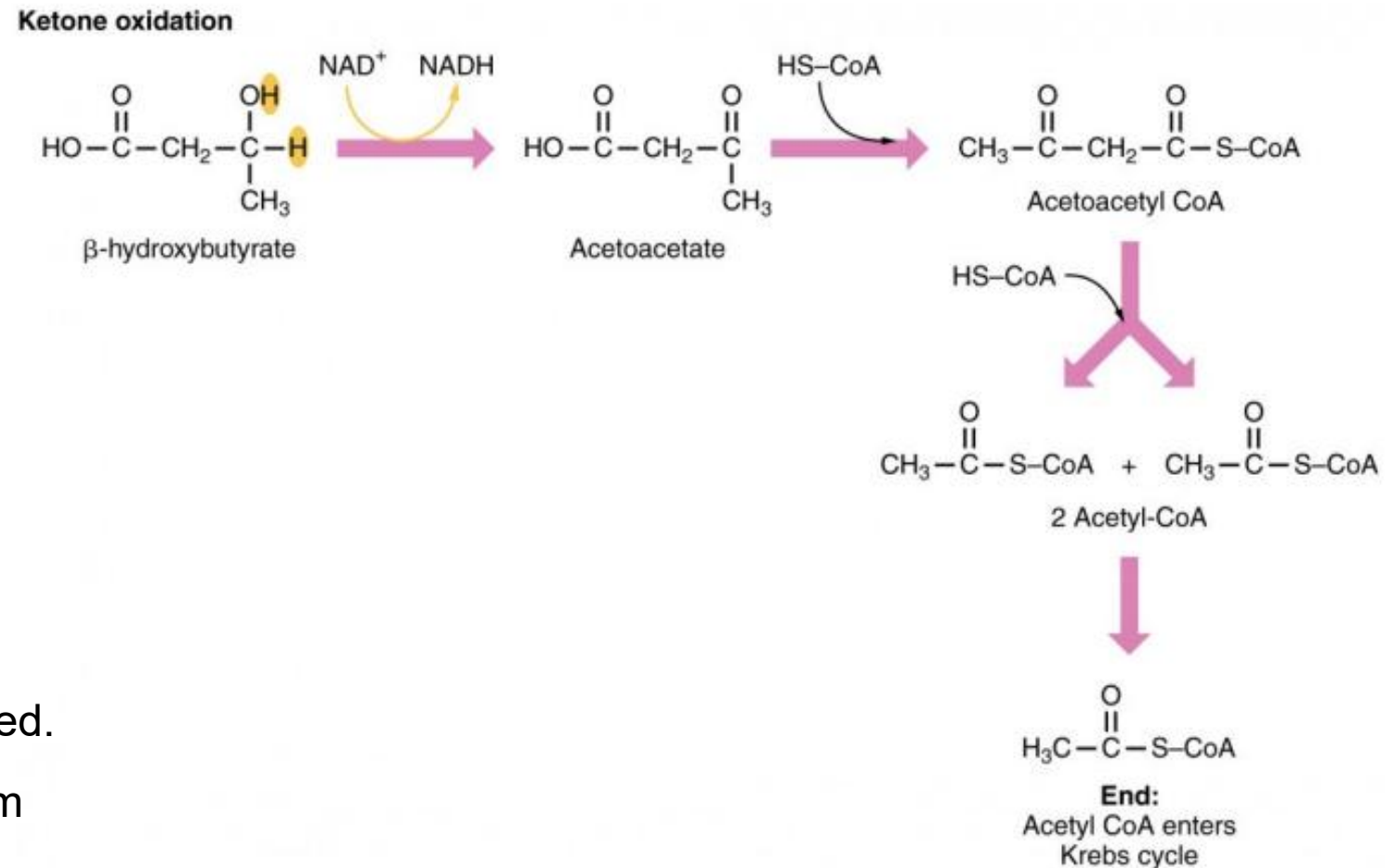
- **Ketogenesis:** Excess acetyl CoA synthesised are converted into ketone bodies (formation of ketone bodies)
- Ketones serve as fuel during prolonged starvation and when the glucose cannot be utilised (energy through Krebs cycle)



Hydroxybutyrate is the primary ketone body found in blood

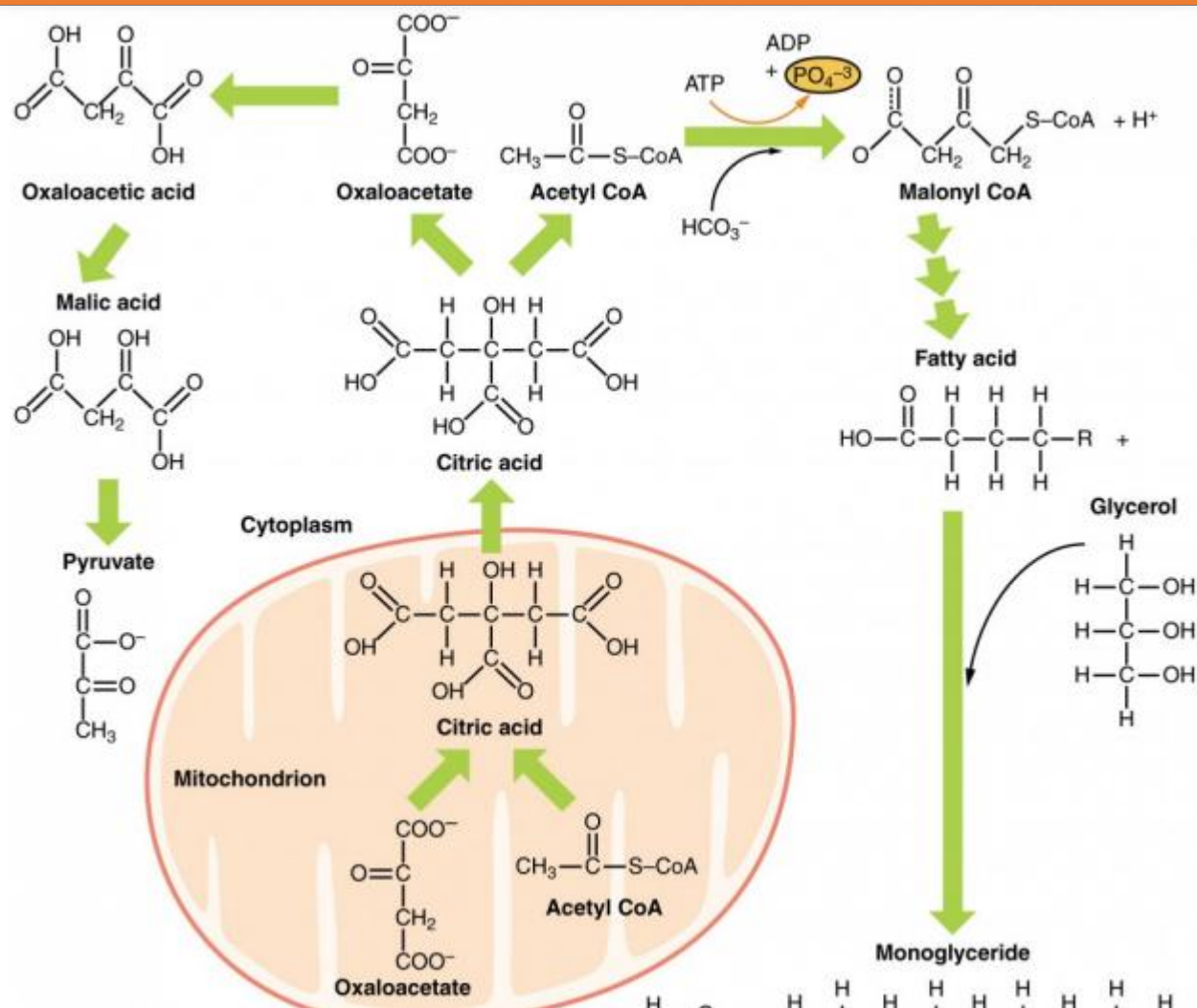
Utilisation of Lipids – Ketone Body Oxidation

- **Ketone oxidation:** Ketone bodies are oxidised to provide energy
- Ketones are the alternate source of energy when the glucose supply is limited – (eg. continuous brain functioning)
- Provide energy by entering into the Krebs cycle
- Breakdown of ketone bodies to carbon-dioxide and acetone when the production of ketone bodies is faster than they can be used. (acetone is exhaled – can be estimated from breath – in diabetic patients)



Utilisation of Lipids – Lipogenesis

- **Lipogenesis:** Synthesis of lipids from acetyl CoA (occurs in cytoplasm)
- When glucose levels are high excess acetyl CoA is converted into fatty acids, triglycerides, cholesterol, steroids, and bile salts
- Process involves addition of two carbons, through acetyl CoA, and proceeds till the required length of the carbon chain is achieved
- Involves transportation of the metabolites across the mitochondrial membrane
- This is an anabolic process and therefore ATP is consumed



In summary ... the 'Power Molecule'

